FitPlan

A Personal Fitness, Workout Planning and Exercise Discovery Web Application

Course	Web Engineering (WE-LAB-F26)
Project type	Semester project
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1. Overview

FitPlan is a client-side, multi-page web application that helps users discover exercises, plan a weekly training schedule, and track completed workouts, all without creating an account.

It combines a public exercise REST API with a curated local JSON dataset, so the app keeps working when the API is unavailable. The project applies core Web Engineering topics, including semantic HTML, responsive CSS, DOM manipulation, asynchronous JavaScript, REST APIs, JSON, browser storage, drag and drop, and timers.

2. Problem Statement

Students and beginner gym users often struggle to keep exercise information, routines, weekly schedules and completed sessions in one place. Existing tools are scattered, filled with ads, or require registration. FitPlan offers a simple, responsive and ad-free alternative that brings discovery, planning, favourites, checklists and workout timing together.

3. Objectives

1. Build a responsive multi-page application using HTML5, CSS3 and JavaScript (ES6+).
2. Integrate a public exercise REST API with a curated local JSON dataset.
3. Let users search and filter exercises by name, muscle group, equipment and difficulty.

4. Provide a 7-day workout planner, favourites with personal notes, and a checklist.
5. Include a guided workout mode with workout and rest timers.
6. Follow good engineering practice, including modular code, Git collaboration, testing and deployment.

4. Scope

In scope	Out of scope
Exercise search, filters and explorers	User accounts and server databases
Exercise details, favourites and notes	Payments or subscriptions
7-day planner with drag and drop	Personal trainer communication
Checklist, timer and guided mode	Medical advice
Progress dashboard and dark mode	Native mobile apps
API plus local JSON, GitHub Pages deployment	Custom backend or cloud database

5. Target Users

Students fitting workouts around classes, beginners who need clear instructions, gym users organising weekly sessions, home workout users with little equipment, and fitness enthusiasts building custom routines.

6. Requirements

Functional

ID	The system shall
FR-01	Let users search exercises by name or keyword.
FR-02	Let users browse by muscle group and category, and filter by equipment and difficulty.
FR-03	Show exercise details, including image, target muscle, equipment, difficulty and instructions.

ID	The system shall
FR-04	Provide an exercise of the day and a random exercise.
FR-05	Let users save favourites and add personal notes.
FR-06	Provide a 7-day planner where users add exercises to specific days.
FR-07	Let users mark exercises as completed using a checklist.
FR-08	Provide workout and rest timers, and a guided workout mode.
FR-09	Merge API results with the local dataset, and fall back to local data when the API fails.
FR-10	Store favourites, notes, plans and checklist data in localStorage.
FR-11	Show loading, empty and error states.
FR-12	Provide a feedback form with client-side validation.

Non-Functional

Attribute	Requirement
Usability and responsiveness	Easy to use on mobile, tablet and desktop.
Performance	Fast page loads with cached API results.
Reliability	Core data stays available offline through the local dataset.
Accessibility	Semantic HTML, alt text, readable contrast and keyboard navigation.
Maintainability	Modular files for API, data, components, pages and utilities.
Security	All user input is sanitised before display.

7. System Design

FitPlan runs entirely in the browser. A central Data Service fetches from the API, merges the local dataset, caches results and handles errors, so pages never call the API directly.

Modules: UI core and theme, exercise search, muscle and equipment explorers, exercise details, favourites and notes, workout planner, checklist, timer, data service, workout categories, progress dashboard, and contact form.

Pages (11): `index`, `explore`, `search`, `exercise`, `workouts`, `saved`, `planner`, `checklist`, `timer`, `progress`, and `about`, each an `.html` file sharing a common header and footer.

8. Technologies and Tools

Area	Choice
Front end	HTML5, CSS3 (Flexbox, Grid, variables), JavaScript ES6+ modules
Browser APIs	Fetch, localStorage, Drag and Drop, timers
Data	Exercise REST API and local JSON
UI	Google Fonts, Font Awesome, Figma or hand sketches
Workflow	VS Code, Git and GitHub, GitHub Pages
Testing	DevTools, W3C Validator, Lighthouse

9. Risks and Mitigation

Risk	Mitigation
API downtime, limits or format changes	Cache responses and fall back to local JSON
API keys exposed in client code	Use a keyless public API, or rely on local data
Drag and drop fails on touch devices	Add an "Add to day" button as an alternative
Unsafe user input (XSS)	Use textContent or sanitise before rendering
Content mistaken for medical advice	Show an informational disclaimer

10. Deliverables

Eleven responsive pages, a curated local JSON dataset, a live GitHub Pages deployment, a GitHub repository with commit history, wireframes and design assets, a test report, and a final report with slides and a live demo.

11. Future Enhancements

User accounts with cloud sync, long-term progress charts, nutrition planning, installable PWA, exercise videos, personalised recommendations, and Urdu language support.

12. Conclusion

FitPlan is a well-scoped front-end project that gives the team practical experience in API and JSON handling, browser storage, responsive design, interactive planning, Git-based teamwork, testing and deployment.